

# Spread Formations: 2x2, Trips, Bunch, Empty, and the Condensed Revolution

**Where the receivers stand, why spreading them across the field makes the defense show its hand, and why the best modern offenses squeezed them back in.**

Third-and-6, midway through the second quarter. The offense breaks the huddle and the running back doesn't stop behind the quarterback. He keeps jogging, out past the right tackle, and settles in a few yards outside the tight end, who is himself standing ten yards from the ball. Nobody is left beside the quarterback. Five players are spread across the width of the field, and the defense starts to move: a linebacker drifts out after the running back, a safety creeps down toward the line, a cornerback points and shouts. The analyst says "empty set, they're going to make the defense show its hand." Two series later the same offense lines up with all three of its wide receivers squeezed in tight, the nearest one barely four yards outside the tackle, and the analyst says "look how condensed they are." Same team, same players, opposite pictures. This chapter explains both: what the spread formations are, why spreading players across the field makes a defense commit, and why so many of the best NFL offenses then pulled their receivers back in.

**You'll be able to...**

- Name the spread formations you'll see every week, 2x2, 3x1 (trips), empty, bunch, stack, nub and quads, in a glance, and number the receivers on each side the way coaches do.
- Explain why spreading receivers across the field forces the defense to declare itself and opens space, and what the offense gives up to get that.
- Count receivers against defenders on each side before the snap and say where the offense has the numbers.
- Explain condensed (tight) splits and why modern offenses use them: crack blocks, the two-way go, rubs, and plays that all look alike.
- Recognize three specialty looks, wildcat, the unbalanced line and the swinging gate, and say what each one is trying to make the defense do.
- Read what the data says about how often NFL teams line up with an empty backfield, and when.

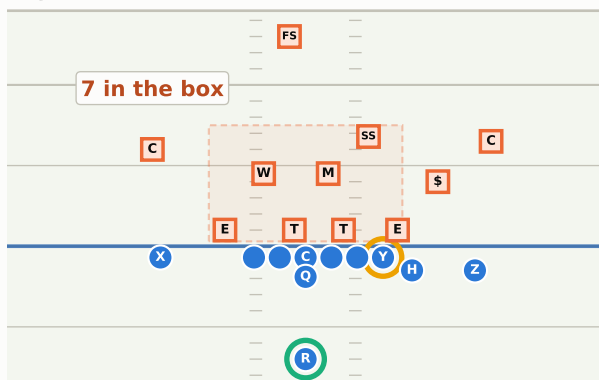
Three ideas from Formation Rules and the Language of Alignment run through this chapter, so here they are in one line each. **Seven players must be on the line of scrimmage**, and only the two at the ends of that line can catch a pass; everyone else is a yard or more behind it, "off the line." **The letters** name spots, not body types: X is the receiver on the line on one side, Z the receiver off the line on the other, Y usually the tight end, H a slot receiver or second back. A split is how far a receiver lines up from the ball or the nearest lineman, and a slot receiver is one who lines up between the outside receiver and the line. And remember the last chapter's rule: personnel is *who* is on the field. Formation, this chapter's subject, is *where they stand*.

## What makes a formation "spread"

A formation is a picture of where the eleven offensive players stand at the snap. For most of football history that picture was compact: the receivers lined up a few yards outside the tackles, the quarterback was under center, and two backs stood behind him. The classic formations, the I, the pro set and the singleback, are all built that way, and they are built to run the ball with as many blockers near it as possible.

A **spread formation** does the opposite. It takes the players who can catch passes and spreads them out across the width of the field, toward both sidelines, usually with the quarterback a few yards back in the shotgun so he can see it all. Nothing about the personnel has to change. The same eleven players can line up either way:

## Tight



## Spread (2x2)

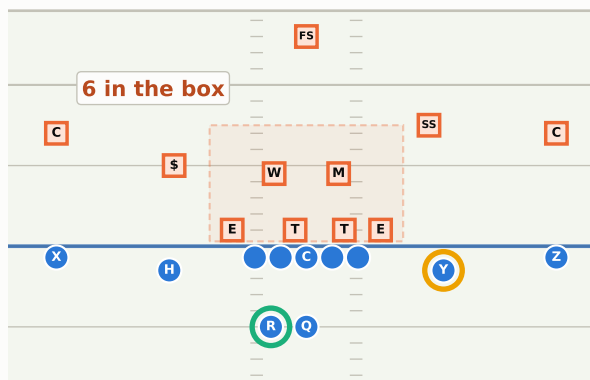


Figure 1: The same 11 personnel (one back, R, aqua ring; one tight end, Y, gold ring; three wide receivers, X, Z and H), lined up two ways against the same nickel defense. Left: a tight formation, receivers close to the line; the defense can keep seven players in the box (shaded: the area near the ball where run defenders stand). Right: a spread 2x2, two receivers on each side out toward the sidelines; defenders have to go with them, and only six are left in the box. Offense: Q = quarterback, C = center. Defense: E = end, T = tackle, M and W = linebackers, \$ = nickel back (the fifth defensive back), C = cornerback, FS and SS = safeties.

Look at what the spread did to the defense without anybody running a step. On the left, the defense can keep seven players in the box: the area near the ball, roughly between the ends of the offensive line and within about seven yards of it, where the defenders who stop runs stand. (The full definition, and the habit of counting the box before every snap, belong to A First Look at Defense.) On the right, the offense has put four players out toward the sidelines. Each of them is a threat to catch a pass, so each needs a defender somewhere near him, and those defenders have to come from somewhere. They come out of the box. Now there are six.

That is the first and most basic reason spread formations exist. **The field is 53½ yards wide, and a spread formation makes the defense defend all of it.** Every defender who leaves the middle to cover a receiver out wide is a defender who isn't there to tackle a runner or to rush the quarterback.

There are three more reasons, and they are what the rest of this chapter is about.

**Spreading out makes the defense declare itself.** When receivers stand close together, defenders can stand close together too, in a clump that hides what they plan to do. When the receivers spread out, the defenders have to spread with them, and where they go is information. A cornerback who follows a receiver all the way across the field is probably playing man-to-man, each defender assigned to one receiver, rather than zone, each defender assigned to an area of the field (both are introduced properly in A First Look at Defense). A safety who walks down toward a slot receiver has told you he isn't going to be deep. A linebacker who stays in the box has told you somebody outside is uncovered or covered by someone farther away. The quarterback reads all of that before the snap. So can you.

**Spreading out creates space.** A receiver near the sideline with one defender in front of him and nobody else within fifteen yards is in a one-on-one race. A quick throw to him, or a run through a box with six defenders instead of eight, gives the ball carrier room to make one man miss.

**Spreading out simplifies the quarterback's job.** With receivers spaced widely, each pass-catcher occupies his own patch of grass, and the quarterback can see who is open by seeing where the defenders *aren't*.

### ! Why it exists

**Spread formations exist to make the defense cover the whole field and tell you how it plans to do it.** They trade blockers near the ball for space away from it. What the offense gives up is the old advantages of tight formations: extra blockers at the point of attack, a quarterback under center who can hand off and fake quickly, and a run game that can simply outnumber the defense. A spread offense can still run, often very well, but it runs by making the box *lighter*, not by bringing more bodies to it.

## Formation, not philosophy

One distinction before we go on. A *spread formation* is a picture at the snap: receivers out wide. A spread offense is a whole philosophy of football, born mostly in college, built on spread formations and on the idea that the field's width is a weapon. Every NFL team uses spread formations every week. Only some of them would call their offense a spread offense. This chapter is about the pictures; the philosophy and its history get their own chapter in *The College Laboratory*, and the systems that grew out of it, like the Air Raid, are installed in Part 4.

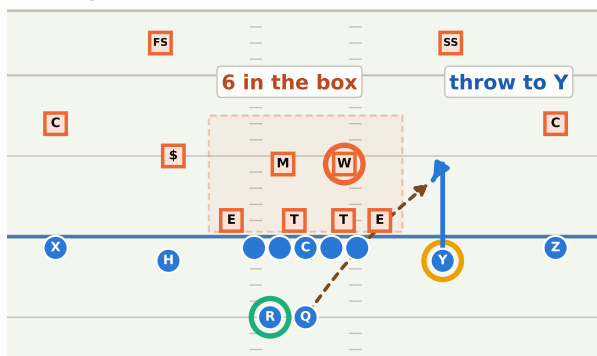
### Common misconception

**"Spread means pass."** Spreading out makes the defense lighter in the box, which is good for *running*. Many college spread offenses were built to run first, and in the NFL, 11-personnel formations with three receivers spread out are only slightly pass-leaning (about 56% pass in neutral situations, as Personnel showed). The one spread formation that really does mean "pass," the empty backfield, is the exception, and we'll measure it below.

## The question every spread formation asks

Spread formations work because they force the defense to answer a question it would rather not answer. Here is the cleanest version of it. The offense lines up in a 2x2 (two receivers on each side). On the right, the slot receiver is the tight end, Y. The defense has a cornerback on each outside receiver, a fifth defensive back (the nickel back, \$) on the left slot, and two safeties deep. Who covers Y?

**W stays inside**



**W walks out**

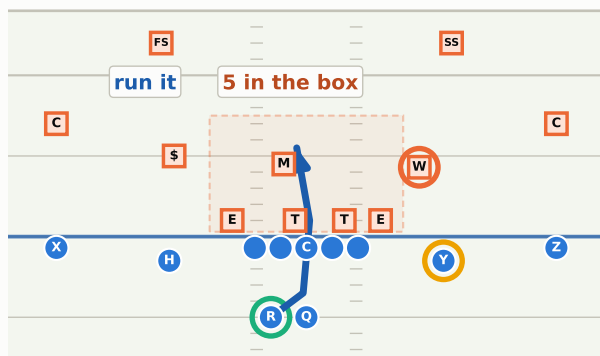


Figure 2: The slot question. Same offense, same defense, one decision: what does the linebacker W do about the slot receiver Y? Left: W stays in the box. The box has six defenders, but Y has nobody within seven yards of him, so the quarterback throws him a quick pass. Right: W walks out over Y to take that throw away. Now the box has five defenders against five offensive linemen, and the offense hands the ball off. Shaded rectangle = the box; the safeties (FS, SS) are deep, 12 yards off the ball.

If the linebacker stays inside, the box is solid against the run but the tight end is alone, and a five-yard throw to him is as easy as a handoff. If he walks out, he takes the throw away and leaves five defenders in the box to face five blockers. Every lineman now has someone to block, and a running back with a one-on-one in the hole is a running back the offense likes. The defense can't be right both ways, because one player can't stand in two places. The offense watches where he stands and does the other thing.

That defender, the one stuck between the box and a slot receiver, is so important that he has his own names, apex and overhang defender, and the passing game builds whole plays around reading him. (The name for a defender with two jobs and time for only one is a conflict defender; you'll meet him again in RPOs, run-pass options in which the quarterback decides after the snap whether to hand off or throw, and which turn this exact question into a play.) For now, notice that the *formation* created the question. In the tight formation there was no slot receiver, so there was no question.

The defense isn't helpless, of course. It can bring a safety down to help, which leaves less help deep. It can play a coverage in which the outside cornerback and the linebacker trade responsibilities. It can disguise its answer and change it at the snap. Those are the defense's moves, and Parts 5 to 7 are full of them. But every one of them costs something somewhere else on the field, and the spread's whole idea is to make the defense pay that cost on every snap.

## i Madden vs. real life

In *Madden*, the defense lines itself up: you pick a play, and the computer slides every defender into place against whatever formation the offense shows. Real defenses have to do that themselves, in about the ten seconds between the offense breaking the huddle and the snap. They do it with **rules agreed in advance**: “against 2x2, the nickel takes the slot to the field and the Will takes the other one; against trips, the safety rotates down.” The middle linebacker and the safeties shout those adjustments to everyone else, and they all have to hear and agree. That’s why offenses love formations the defense hasn’t prepared for, and why a defense that sees a formation it didn’t practice sometimes ends up with two players covering one receiver and nobody on another.

## Counting receivers: #1, #2, #3

Before we name formations, you need the coaches’ counting system, because every formation name and every coverage rule is built on it. **Receiver numbering** counts the receivers from the sideline in, separately on each side of the ball. The widest receiver on a side is #1, the next one in is #2, the next is #3. It doesn’t matter whether the receiver is a wide receiver, a tight end or a running back who has lined up out wide: if he’s out of the backfield, he gets a number. A back standing beside or behind the quarterback isn’t counted until he leaves.

Why number them instead of using names? Because the defense doesn’t care who the receivers are; it cares where they are. A coverage rule like “the nickel back takes #2” works against any formation and any personnel. The offense can swap which players stand where, and the rule still tells the nickel whom to cover. The numbers also give a formation its name: two on each side is “2x2,” three on one side and one on the other is “3x1.” The side with the bigger number is the formation’s *receiver strength* (or passing strength), the side with more threats to catch the ball, a cousin of the strong side set by the tight end.

Here are the four formations you’ll see most, all with the same 11 personnel:

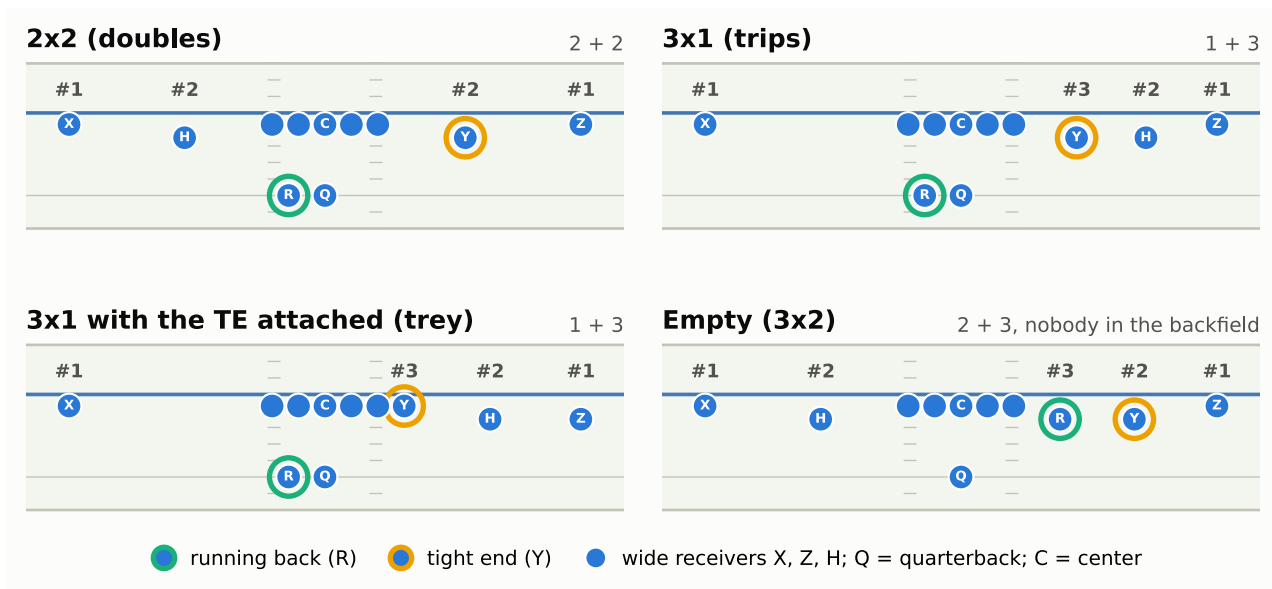


Figure 3: The core spread formations, all from the same 11 personnel (one back, R; one tight end, Y; three wide receivers, X, Z and H) with the quarterback (Q) in the shotgun. The numbers count each side’s receivers from the sideline in; the back beside the quarterback isn’t counted. Top left: 2x2, two on each side. Top right: 3x1 (trips), three detached receivers on the right and X alone on the left. Bottom left: the same 3x1 count with the tight end attached to the line, which many teams call ‘trey’ instead of trips.

Bottom right: empty, the back has left the backfield, so all five receivers are out, three to one side and two to the other (3x2).

The names vary by team. One staff’s “Doubles” is another’s “Deuce” or “Spread”; “Trips” is also “Trio,” “Tre” or “Triangle,” and whether the tight end attached to the line makes it “trips” or “trey” depends on whose playbook you’re reading. The counting doesn’t vary, which is why this book uses the numbers.

## Common misconception

**“Trips means three wide receivers.”** It means three *receivers* on one side, whatever their position. In the plate above, the third receiver in trips is a tight end, and in empty the third man on the right is the running back. Personnel counts bodies; formations count *spots*. When the broadcast says “11 personnel, trips right,” that’s two separate facts: who is on the field, and where three of them are standing.

## 2x2: balance

A 2x2 formation, also called **doubles**, has two receivers on each side of the ball. In the most common NFL version the outside receivers (#1s, X and Z) stand on the line near the sidelines and the inside receivers (#2s, the slots) stand off the line, a few yards inside them. The slots *must* be off the line: if a slot receiver stood on the line, the outside receiver would be “covered” (no longer at the end of the line) and couldn’t legally catch a pass. That’s the covered receiver rule, and it’s why you’ll almost always see a slot receiver a step behind the receiver outside him.

The 2x2’s virtue is **balance**. The defense can’t tell from the alignment which side the play is going to, because the two sides look alike. The offense can run the same route combination to both sides, called a “mirrored” concept, and let the quarterback pick whichever side the defense covers worse. Or it can run two *different* combinations and pick based on what the defense shows. Against a defense with two deep safeties, a 2x2 asks each safety to help on his own side, two receivers apiece, and a defense that wants to bring an extra defender into the box has to decide which side gets less help.

What 2x2 gives up is **concentration**. Because the formation is balanced, the defense can be balanced too: it can play the same coverage to both sides with the same number of defenders, and nothing about the alignment forces it to shift. The formations that follow all trade some of that balance for an advantage somewhere specific.

## 3x1: the overload

Put three receivers on one side and one on the other, and you have a 3x1 formation, almost universally called **trips**. It’s the formation of the opening play in *What a Play Really Is*, “Gun Trips Right,” and it’s one of the most common pictures in the NFL.

Trips does two things at once. On the three-receiver side, the *trips side*, it puts more receivers than the defense normally has defenders, so the defense has to send help. On the other side, the *backside*, it leaves one receiver, often the offense’s best, alone with one cornerback. The defense has to choose which of those problems to solve:

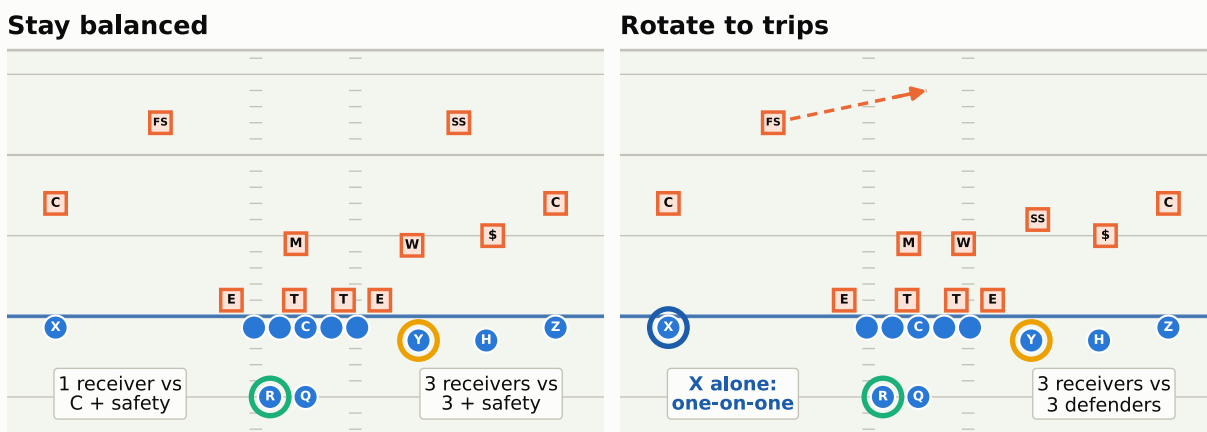


Figure 4: Trips makes the defense choose. Left: the defense stays balanced with two deep safeties. The three receivers on the right face three defenders (the cornerback C, the nickel back \$ and the linebacker W) plus a deep safety, and the lone receiver X on the left has his cornerback and a safety over the top. Right: the defense rotates toward trips. The left safety (FS) moves to the deep middle and the right safety (SS) comes down over the third receiver, so the trips side gets an extra defender, but X is now one-on-one with no safety near him. Either way the defense has given something up.

If the defense stays balanced, the trips side is three receivers against three defenders underneath and one safety, and the offense can attack it with combinations designed to put two receivers on one defender. If the defense rotates, it gets an extra man to the trips side, but the backside receiver is now alone on what coaches call an *island*, one-on-one with nobody behind his cornerback.

That's why offenses so often put their best receiver at the single-receiver spot in 3x1. They're betting he wins one-on-one, and if the defense refuses that bet by keeping a safety on his side, the trips side is short-handed.

The trade-off for the offense is that **trips announces its strength**. The defense knows which side has the numbers and can tilt its coverage, its front and its blitzes (extra pass rushers; see A First Look at Defense) toward it. Trips also leaves the backside thin for the run game: with only one receiver over there, the defense can squeeze its run defenders toward the trips side without worrying much about a quick throw the other way. Offenses answer by running *away* from trips, toward the single receiver, where the defense has thinned out; the run game chapters show how.

Trips also interacts with the hash. With the ball on a hash mark, one side of the field is about six yards wider than the other (the field side and the boundary side). Most offenses put trips to the field side, where three receivers have room to spread out, and leave the single receiver to the short side. When an offense puts trips into the *boundary*, squeezing three receivers into the narrow side, it's usually a deliberate surprise: the defense's rules for the field and the boundary may not fit.

## Empty: everyone out

An **empty** formation has nobody in the backfield but the quarterback. All five eligible receivers are out, most often three on one side and two on the other, so it's also called **3x2** (and "5-wide" on broadcasts). As the plates above showed, empty usually isn't a special personnel grouping. The running back and tight end simply line up as receivers. In 2025, about 72% of NFL empty snaps came from ordinary 11 personnel.<sup>1</sup>

Empty is the spread taken to its limit, and it does three things nothing else does.

**It shows the quarterback everything.** With five receivers spread out, the defense has to put someone near each of them, and the quarterback can often make a good guess at man coverage or zone before the snap just by watching who goes where. If a linebacker runs out with the running back, that's man. If he stays inside and lets a safety take the back, that's probably zone.

**It isolates the backs and tight ends.** A running back split out wide is often covered by a linebacker, who is bigger and slower than the defensive backs. If the defense brings in more defensive backs to cover him, it is smaller everywhere else.

**It exposes the blitz.** Five receivers out means only the five offensive linemen are left to block. If the defense shows six pass rushers, everyone in the stadium can count: five blockers, six rushers, one man free. The offense knows it in advance, and so does the quarterback, so the answer is built into the play: get the ball out before the free man arrives.

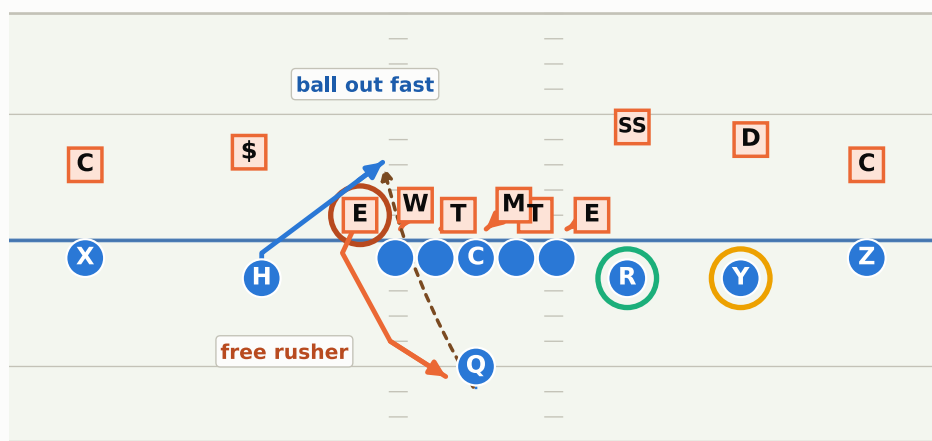


Figure 5: Empty against a six-man pressure. The defense lines up six players at the line of scrimmage (four linemen and both linebackers, M and W) and plays man-to-man with no deep safety. The five offensive linemen block five of the rushers; the sixth, the end on the left (ringed), comes free. The quarterback doesn't wait for him: H runs a quick slant into the space the blitzing linebacker just left, and the ball comes out within about a second of the snap. With five receivers and five blockers, every extra rusher the defense sends is one fewer defender in coverage.

The trade-offs are just as clear. Empty leaves nobody to pick up a blitz, so the quarterback has to throw quickly and accurately every time the defense brings pressure, and some of those throws are short of the first-down marker. It makes the run game nearly

<sup>1</sup>Course calculation from nflverse data: 2025 regular-season runs and passes with no offensive player in the backfield (FTN charting), joined to the participation file's personnel. 11 personnel made up 71.7% of empty snaps, 12 personnel 14.9%, and every other grouping 3% or less.

disappear: there's no back to hand off to, so the only run left is the quarterback himself. And it is the most **predictable** formation in the spread family, because everyone knows what it's for.

How predictable? Here is the empty-backfield rate in the NFL over the last decade, measured two ways, and what offenses did from it:

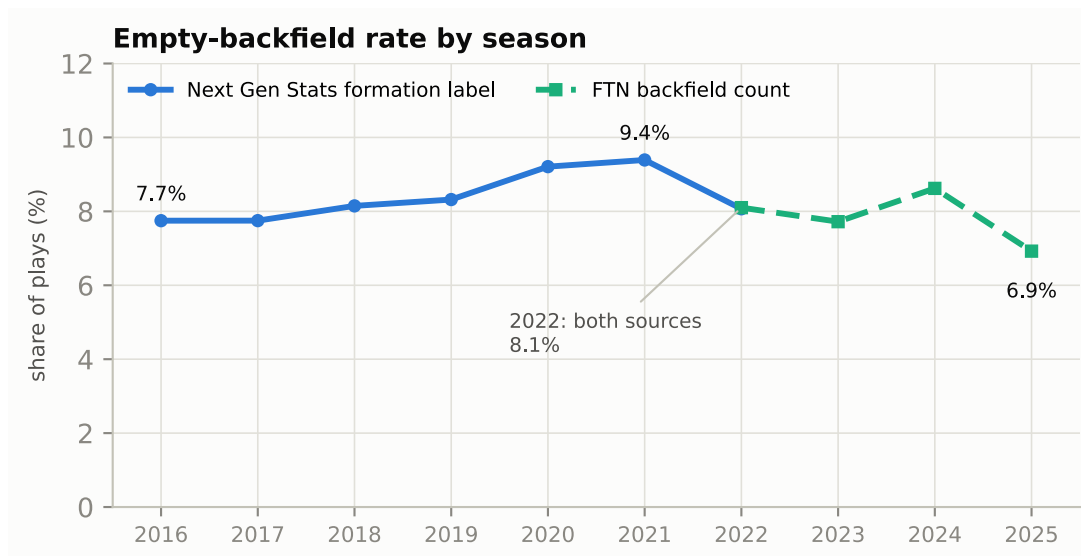


Figure 6: How often NFL offenses lined up with an empty backfield, regular-season runs and passes, 2016–2025 (course calculation from nflverse data). Through 2022 the formation label comes from Next Gen Stats; from 2022 on, from FTN’s charted count of players in the backfield. The two sources overlap in 2022 and agree (8.1% each), so the series can be read as one. Empty has stayed between about 7% and 9.5% of plays for a decade: a specialty tool, not a base formation.

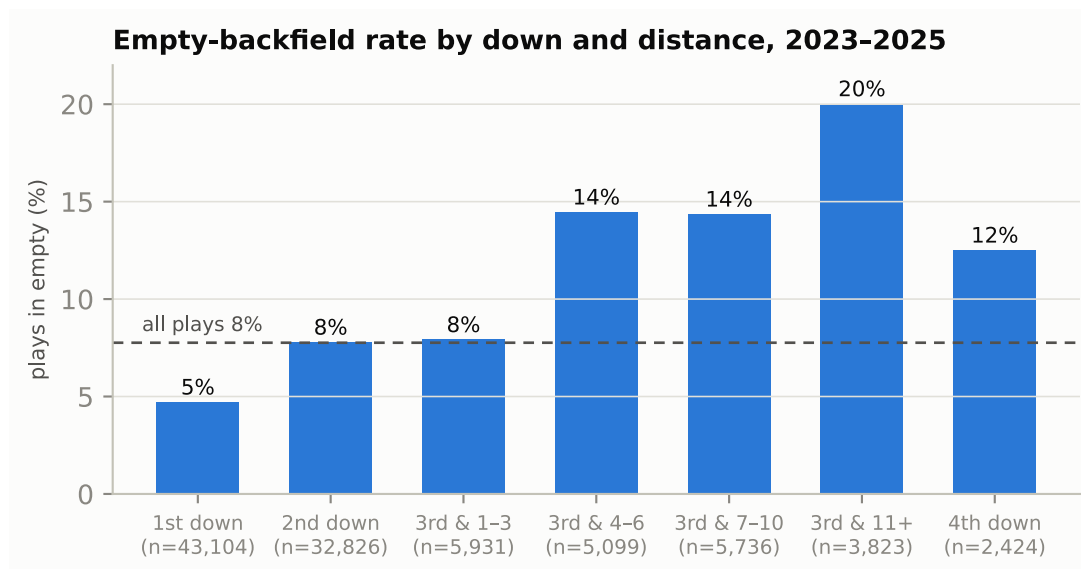


Figure 7: When offenses go empty: the share of plays with an empty backfield by down and distance, 2023–2025 regular seasons (FTN charting via nflverse). Empty is rare on first down and most common on third down with more than ten yards to go, the down and distance on which the offense most needs a completion and the defense is most likely to blitz. The dashed line is the rate across all plays.

Two numbers tell the story. First, empty is a *specialty* formation: between about 7 and 9.5 percent of plays, every season for a decade, with no big trend either way. Second, it is overwhelmingly a passing formation: in 2023–2025, more than 94% of NFL plays from empty were dropbacks, against about 58% from everything else.<sup>2</sup> Offenses save it for third down, and especially third-and-long, when they need a completion and expect the defense to blitz.

<sup>2</sup>Course calculation from nflverse data (`load_pbp`, `load_participation`, `load_ftn_charting`), regular-season plays with `play_type` of pass or run and a recorded down. Empty through 2022 is the Next Gen Stats `offense_formation` label “EMPTY” (the fewer than 1% of plays with no formation label are left out); from 2022 it is FTN’s `n_offense_backfield` equal to zero. In 2022, the one season both sources cover, they agree at 8.1%. A pass is any dropback,

That predictability is the price, and it's why empty is a *tool* rather than a base formation. If empty is almost always a pass, the defense can rush the passer without worrying about the run. Offenses that use it a lot find ways to make it less predictable, and the best way is a quarterback who is dangerous running the ball: with a running quarterback, five defenders in the box against five blockers is a fair fight for *him*. You'll see that in the film room with the Buffalo Bills.

**i** Go deeper: why there's no free 2x2-versus-3x1 chart

You might expect a chart of how often teams use 2x2 and 3x1 too. The free play-by-play data doesn't record it. Next Gen Stats' formation labels (2016–2022) say only SHOTGUN, SINGLEBACK, I\_FORM, EMPTY and the like, and from 2023 the labels shrink to just SHOTGUN, UNDER CENTER and PISTOL. FTN's charting counts players in the backfield but not where the receivers stand. That's why the empty chart uses two sources: the only "spread" formation both of them can see is the one with nobody in the backfield.

The NFL's **Big Data Bowl** files do record it. The 2025 edition (2022 season, weeks 1–9) includes a `receiverAlignment` label for every play ("2x2," "3x1," "3x2," "2x1" and so on) and the tracking coordinates to check it yourself. The data lives on Kaggle (a data-science competition site) and needs a free login, so the code for this, in the web edition of this chapter under "Show the Python," is not run when the book is built.

## Clusters: bunch and stack

Spreading receivers apart is one idea. Putting them *close together* on purpose is another, and it's just as old. Here are the four variations you'll see most, again all from one personnel group:

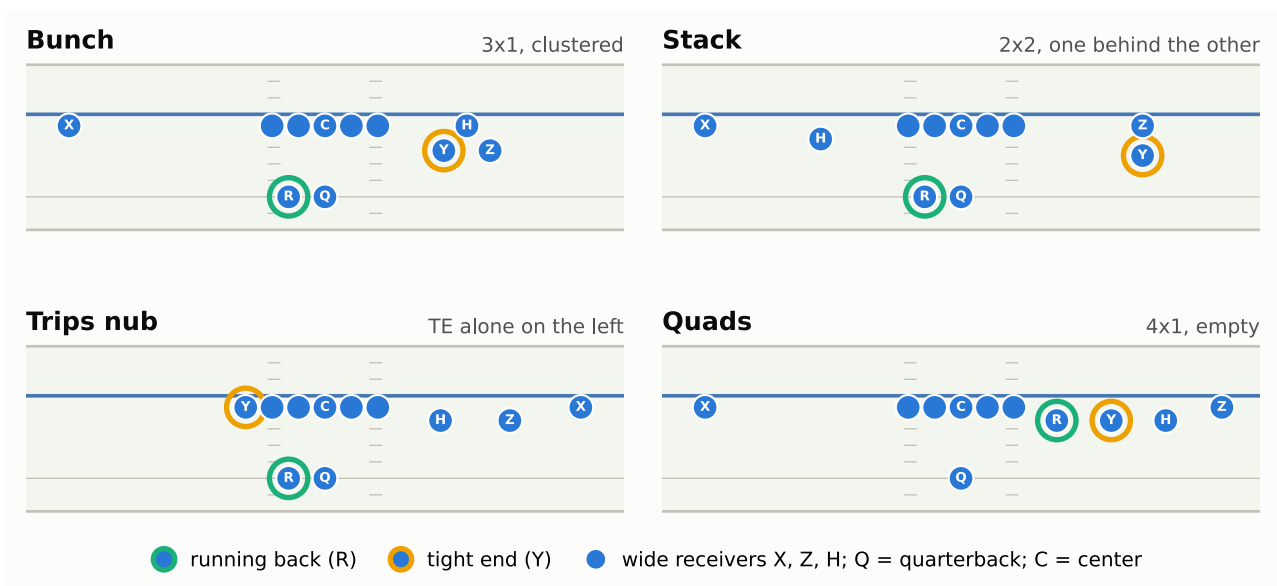


Figure 8: Four variations, again all 11 personnel. Top left: bunch, three receivers in a tight triangle on the right (H on the line, Y and Z a yard or two behind and either side of him). Top right: stack, two receivers one directly behind the other (Z on the line, Y behind him). Bottom left: trips nub, three receivers to the right and only the tight end on the left, attached to the line with nobody outside him. Bottom right: quads, four receivers on the right (the back, R, among them) and X alone on the left, with nobody beside the quarterback.

A **bunch** is three receivers clustered in a tight triangle, usually within two or three yards of each other: one on the line (the "point"), the other two just off it, one inside and one outside. A **stack** is two receivers lined up one directly behind the other, the front man on the line and the back man a yard or two behind. Both are legal for the same reason as everything else in this chapter: only one receiver in each cluster is on the line, and he's the end of it.

Why would an offense that wants space squeeze its receivers together? Three reasons, and they all have to do with defenses that play man-to-man.

including sacks and scrambles. Data dictionaries: [https://nflreadr.nflverse.com/articles/dictionary\\_participation.html](https://nflreadr.nflverse.com/articles/dictionary_participation.html) and [https://nflreadr.nflverse.com/articles/dictionary\\_ftn\\_charting.html](https://nflreadr.nflverse.com/articles/dictionary_ftn_charting.html).

**Traffic.** In man coverage, each defender follows one receiver wherever he goes. When three receivers start within two yards of each other and then run in different directions, their paths cross, and the defenders chasing them have to get around each other and around the other receivers. A defender who is a step late is a receiver who is a step open. Watch it happen:

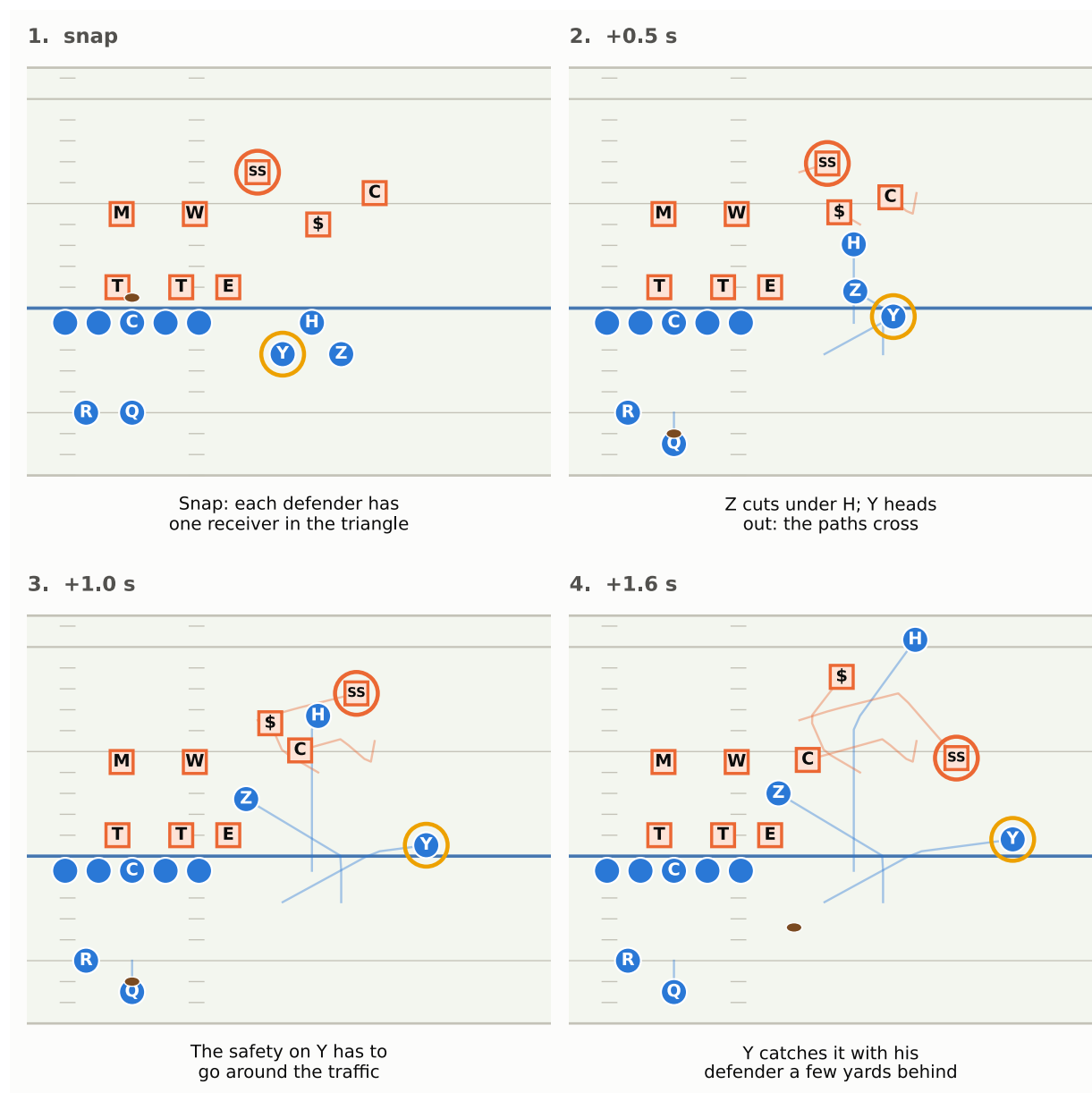


Figure 9: Bunch against man coverage (simulated, BDB-format tracking). 1: each receiver in the bunch has a defender assigned to him: the cornerback (C) on Z, the nickel (\$) on H, the safety (SS) on the tight end Y. 2: at the snap Z cuts inside under H while Y heads for the flat, so their paths cross right in front of the defenders. 3: the safety covering Y has to work around the crossing receivers. 4: Y catches the ball in the flat with his defender still a few yards behind. Nobody blocked anybody; the alignment did the work.

The receivers never touched the defenders, and that matters. Running into a defender on purpose to knock him off a receiver, after the snap and before the pass is touched, more than a yard past the line of scrimmage, is offensive pass interference.<sup>3</sup> So bunch and stack plays are designed so that the *receivers' own paths* get in the defenders' way, or so that a receiver who runs his route exactly as drawn happens to cross in front of a defender. Coaches call that a “natural pick” or a “rub.” The full family of rub routes is in Dropback Concepts.

<sup>3</sup>NFL Rulebook, Rule 8, Section 5 (pass interference): pass interference is an act more than one yard beyond the line of scrimmage that significantly hinders an eligible player's opportunity to catch the ball; for the offense the restrictions run from the snap until the ball is touched, and “blocking downfield by an offensive player prior to the ball being touched” (the pick play) is the first listed example of offensive pass interference: <https://operations.nfl.com/the-rules/nfl-rulebook/>.

**Free releases.** A cornerback who wants to jam a receiver at the line (to press him) has to line up right in front of him. In a stack, only the front receiver can be pressed; the man behind him starts with a free run. In a bunch, the two back receivers are shielded in the same way. That makes clusters a favorite answer to defenses that press.

**Communication.** Clustered receivers force defenders to decide, at the snap, who takes whom. Many defenses don't try to follow their man through the traffic; they agree to *swap*, so the defender nearest the receiver going out takes him and the defender nearest the receiver going in takes that one. Swapping works, but it takes two defenders who see the same thing at the same time and talk. Every time a defense has to communicate under stress, it can get it wrong.

What clusters give up is **width**. Three receivers within three yards of each other can be covered by fewer zone defenders than three receivers spread across the field, because a zone defender standing in the right spot can cover two of them. That's why bunch and stack are at their best against man coverage, and why you'll see them most in the red zone and on third-and-short, where defenses play the most man.

## Lopsided: nub and quads

Two more names describe formations by what's *missing*.

A **nub** side is a side of the formation where a tight end is the only receiver, attached to the line with no wide receiver outside him. "Trips nub" (the bottom-left plate above) puts three receivers to one side and only the tight end on the other. It's a clever shape, because it splits the formation's identity in two. The tight end side is the **strong side** for the run game (it has the extra blocker), while the trips side is the receiver strength (it has the extra pass-catchers). A defense that sets its front to the tight end is undermanned against the receivers, and a defense that rotates its coverage to the receivers is light against a run to the tight end. The tight end is also hard to cover from that spot: there's no wide receiver to put a cornerback on, so the defense often has a linebacker or an end responsible for him, and he can block either of them or release into a route.

**Quads** puts four receivers on one side. Since only five players can be eligible, quads leaves either one receiver on the other side and nobody beside the quarterback (the "empty quads," or 4x1, shown above), or a back beside the quarterback and nobody at all on the other side. It's the most extreme overload in football, and it's rare: you'll see it in two-minute drills, on trick plays, and from offenses testing whether the defense has a rule for it. Its value is the question it asks. A defense that hasn't practiced against four receivers on one side has to improvise, and improvised coverage is how receivers get open.

## The condensed revolution

Everything so far might suggest the answer is always "spread out more." For about a decade, that was roughly where NFL offenses were heading. Then, in the late 2010s, some of the most successful offenses in the league started doing something that looked like going backward: they pulled their receivers *in*.

A **condensed formation** lines its receivers up close to the offensive line, with reduced splits (inside the painted numbers) or even tight ones (a few yards outside the tackle or tight end), instead of out near the sidelines. The personnel is the same; the receivers just stand closer to the ball. Sean McVay's Los Angeles Rams made it their signature in the late 2010s, Kyle Shanahan's San Francisco 49ers took it furthest, and much of the league followed: by Next Gen Stats' count, condensed alignments went from about 11% of NFL plays in 2016 to outnumbering spread ones in 2023.<sup>4</sup>

Why squeeze back in? Because a receiver who stands close to the line can do things a receiver near the sideline can't. There are four of them.

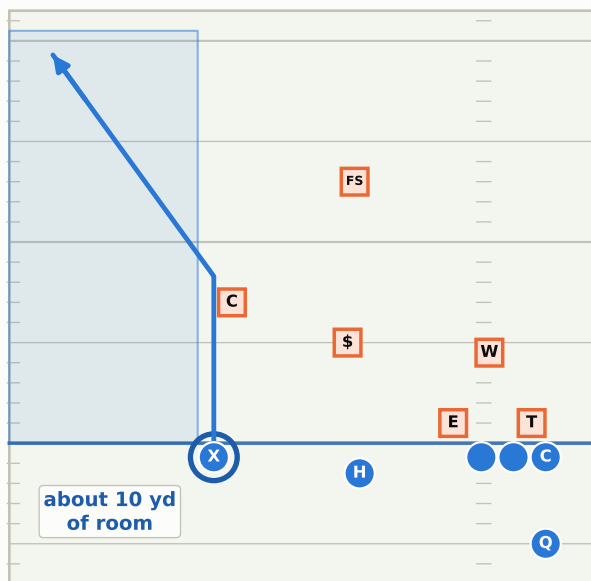
### 1. Room to the outside

Start with the simplest. A receiver lined up near the sideline has nowhere to go outside him. The sideline is out of bounds, so any route that breaks outward runs out of room, and the cornerback covering him knows it. He can line up inside the receiver and let the sideline guard the outside, like a twelfth defender. Move the receiver in toward the ball and all that grass is suddenly his:

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<sup>4</sup>Domonique Foxworth, "Yes, Sean McVay is an offensive genius," *Andscape*, Jan. 29, 2019 (Rams' condensed formations): <https://andscape.com/features/rams-coach-sean-mcvay-is-an-offensive-genius>; Zach Barnett, "How Kyle Shanahan has mastered the play-calling trend that's sweeping the NFL," *FootballScoop*, Feb. 6, 2024, summarizing a *Wall Street Journal* analysis of Next Gen Stats data (condensed about 11% of plays in 2016; tight splits overtook the spread in 2023; the 49ers' formations the narrowest in the league, about 19.9 yards end to end): <https://footballscoop.com/news/how-kyle-shanahan-has-mastered-the-play-calling-trend-thats-sweeping-the-nfl>.

### Wide split



### Condensed split

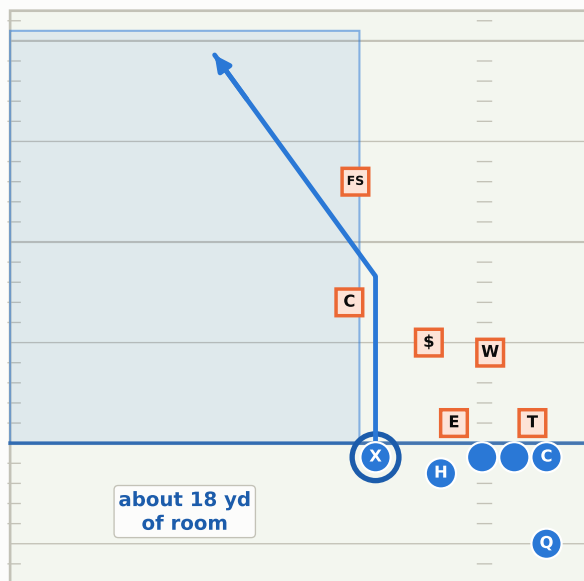
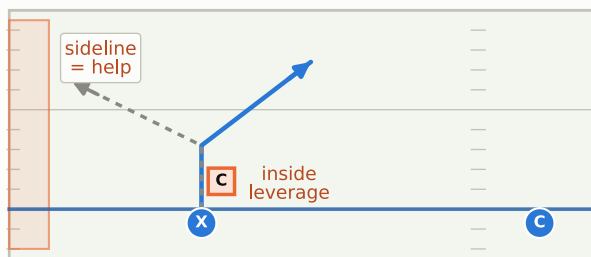


Figure 10: The same route from two splits, left half of the field. Left: X lines up wide, about ten yards from the sideline. His corner route (straight up, then breaking toward the sideline) has little room, and the cornerback (C) can line up inside him and let the sideline do the rest. Right: the same receiver condensed, about eight and a half yards from the ball. Now there are about 18 yards of grass outside him (shaded), the corner route has room to run, and the cornerback has to move outside to protect that space, which opens the inside for routes that cut toward the middle. The space outside the receiver is shaded in both panels.

## 2. The two-way go

That leads to the second reason. A receiver with room on *both* sides of his defender has what coaches call a **two-way go**: he can break inside or outside, and the defender has to guess. Against a receiver near the sideline, the defender doesn't guess; he takes away the inside and lets the sideline take away the rest. Against a receiver with a two-way go, a defender who leans one way gets beaten the other.

### Near the sideline: one way



### Room both ways: two-way go

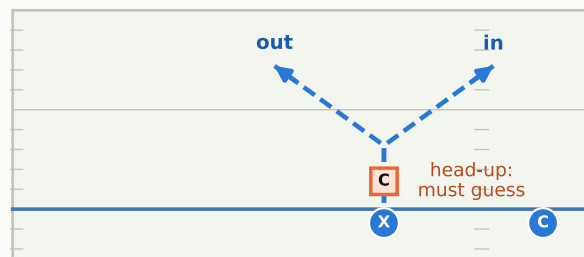


Figure 11: The two-way go. Left: a receiver near the sideline. His defender lines up on his inside shoulder (inside leverage), because the sideline protects the outside; the receiver's only real option is to fight back inside. Right: a receiver lined up in the slot or with a condensed split. With room on both sides, his defender lines up head-up and has to guess; the receiver can break in or out (both dotted options), and whichever way the defender leans, the receiver goes the other.

The defender's position relative to the receiver, inside him, outside him or straight in front, is called his leverage, and the quick passing game is largely a contest of reading it. The point here is about the formation: **where a receiver lines up decides how many ways he can go, and so how much the defender has to guess**. Slot receivers have always had two-way gos. Condensed splits give one to the outside receivers too.

## 3. Crack blocks

The third reason is the run game. A receiver who lines up close to the formation is *already inside* the defenders who would normally be his problem, and that lets him block them from the side. A crack block is a wide receiver blocking back toward the ball, on a linebacker or safety who is watching the play and not him. It's one of the most effective blocks in football, because the defender

is looking the other way and the receiver gets a running start. From a wide split it takes too long: by the time the receiver gets back inside, the play is over. From a condensed split he's there in a step or two. Here's the classic version, a toss (the quarterback pitches the ball underhand to a back who is already running wide) with the receiver cracking inside and a pulling lineman (one who leaves his spot at the snap and runs along the line to block somewhere else) replacing him on the cornerback:

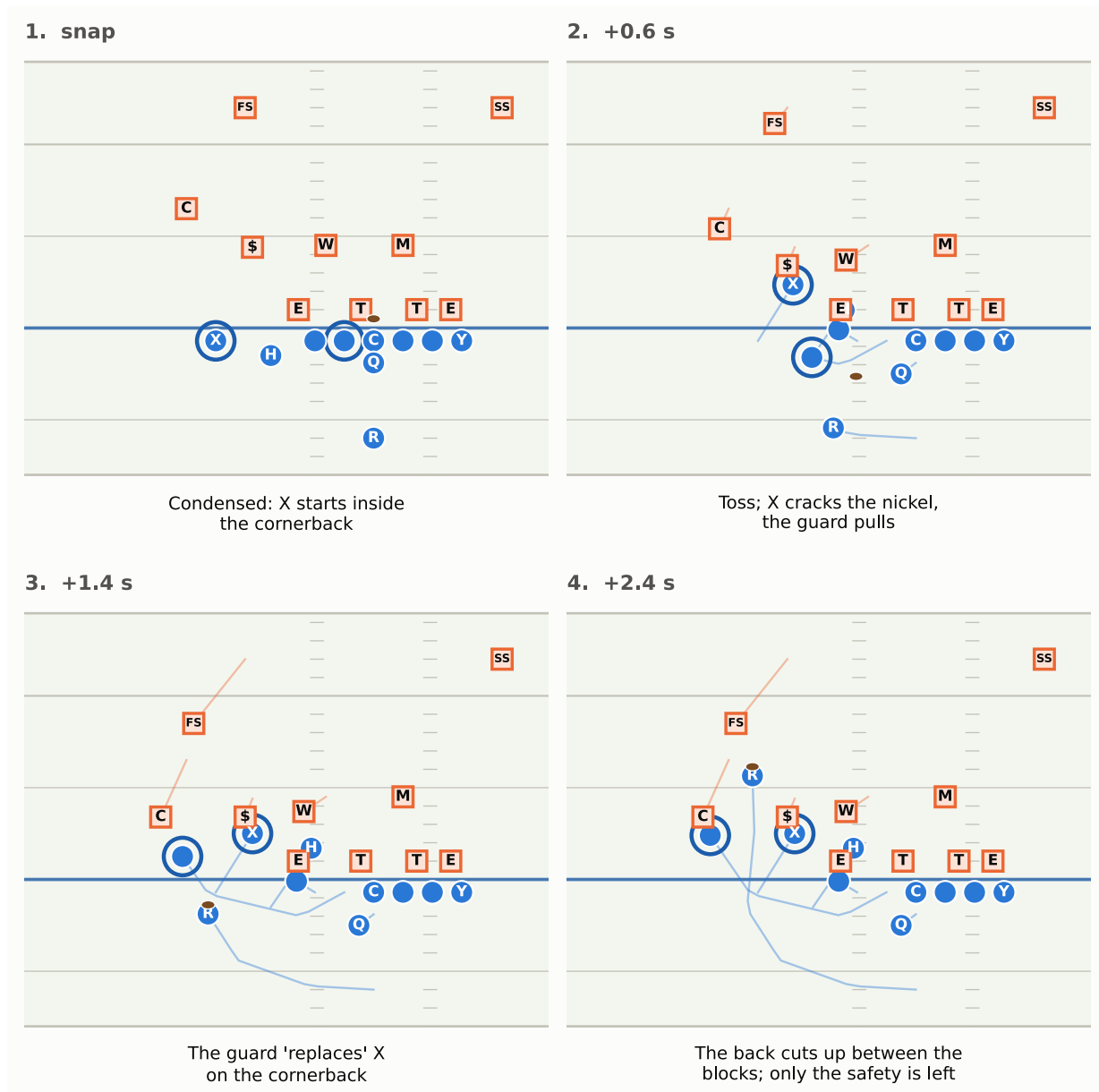


Figure 12: Crack toss from a condensed formation (simulated, BDB-format tracking). 1: X lines up only about eight and a half yards from the ball, inside the cornerback. 2: the quarterback tosses the ball to the running back (R) heading left; X blocks down (cracks) on the nickel back (\$) and the left guard pulls around the end. 3: the guard blocks the cornerback, the man X would have blocked from a wide split; H blocks the linebacker. 4: the back turns upfield between X's crack block and the guard's block, with only the deep safety left to beat. The ringed players are the two whose jobs the condensed split makes possible.

Notice the defender who got blocked. The nickel back was lined up to stop the toss, watching the ball, and the receiver he was "covering" turned around and hit him from the side. If the nickel starts watching the receiver instead, the receiver runs past him on a pass. That's the condensed formation's version of the slot question from earlier in the chapter: a defender who has to worry about the receiver *and* the run can't do both. The perimeter run game, with its toss sweeps, jet sweeps and crack-and-replace blocking, is in Perimeter Runs and Misdirection.

#### 4. Everything looks the same

The last reason is the subtlest, and for many coaches the most important. When receivers stand close to the formation on every play, a run and a pass start from **the same picture**. The receiver who cracks on a run is lined up exactly where he lines up to

run a route on play-action (faking the handoff, then throwing). The defense can't read the play from the splits. The condensed formations of the late-2010s Rams and 49ers were built around exactly that: outside zone runs (on which the whole line steps sideways together and the back aims for the edge), crack blocks and play-action passes, all from one look, so a linebacker or safety who guessed "run" because the receivers were tight was guessing wrong half the time.

## What condensed gives up

Nothing in football is free, and condensing has two clear costs.

**It brings the defense in too.** Defenders line up on receivers, so when the receivers move in, the defenders move in, and the box can get *more* crowded, not less. The spread's first benefit, thinning the box, runs backward. Condensed offenses accept that because the receivers now block those extra defenders, but a condensed formation against a defense that plays it well can feel very crowded.

**It gives up horizontal stretch.** A spread formation makes defenders cover sideline to sideline; a condensed one lets them stay closer together, so one defender can sometimes cover two receivers, and quick throws to the outside, like screens to a receiver near the sideline, have less room. The horizontal stretch, making defenders cover too much width, is one of the passing game's basic weapons, and condensing trades some of it for the other advantages.

### ! Why it exists

**Condensed formations exist because the spread taught defenses to play wide.** Once defenses learned to spread out and play the receivers from the inside, using the sideline as help, the open space moved: it was now *outside* the receivers and in the run fits near the ball. Condensing puts the receivers where the new space is, and where they can block the defenders who are watching the ball. It trades width for angles. It's the same idea as the spread, making the defense defend space it would rather not, aimed at a different patch of grass.

## Specialty looks

Three more formations deserve names, not because you'll see them every drive but because when you do see them, they're asking a specific question, and you can answer it before the snap.

### Wildcat: a direct snap to someone else

In the **wildcat**, the ball is snapped directly to a running back (or a receiver, or a tight end) standing where the shotgun quarterback would be, while the real quarterback lines up as a receiver or leaves the field. Usually another back or receiver runs in motion across the formation at the snap, so the player who takes the snap can hand it to him, fake to him, or keep it himself.

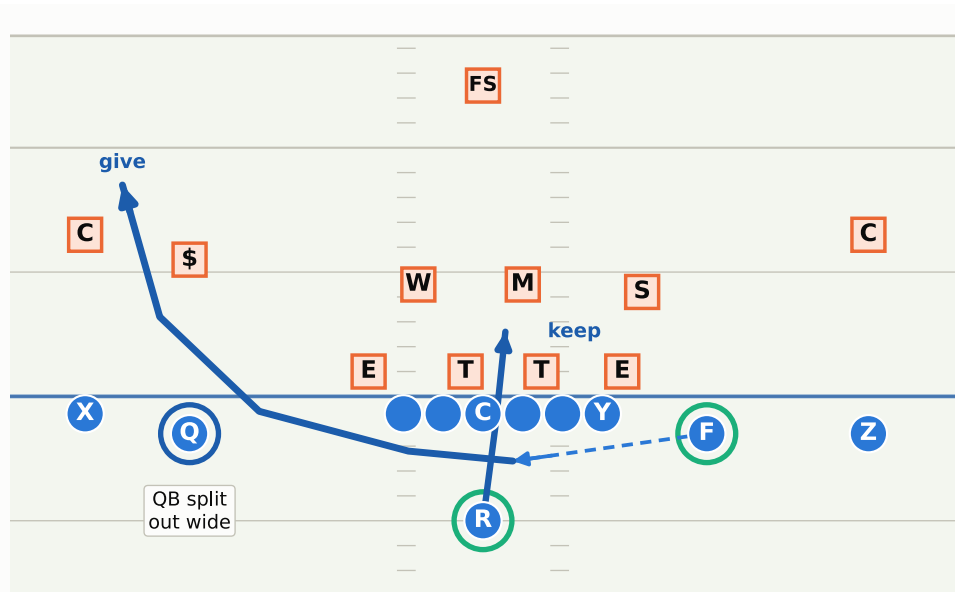


Figure 13: A typical wildcat. The running back (R) stands in the shotgun spot and takes the snap; the quarterback (Q) is split out on the left as a receiver. A second back (F) runs in motion from the right (dashed) and crosses in front of R at the snap: R either hands it to him on a sweep to the left (give) or keeps it and runs inside (keep). The defense must still put someone on the quarterback out wide, so it has one fewer defender for a run play in which every blocker is accounted for.

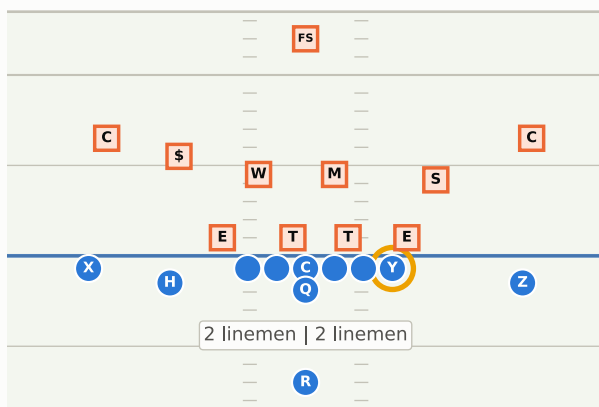
Why snap to a running back? Think about a normal handoff. The quarterback takes the snap, hands the ball off, and then does nothing useful for the run: he's one of the offense's eleven players, but he isn't blocking anybody. The defense, meanwhile, has all eleven of its players available to tackle. In the wildcat, the player who takes the snap is himself a running threat, and the quarterback, now split out wide, still needs a defender on him. So the offense has one more player who matters to the run and the defense has one fewer to stop it. The motion man adds a second runner, so the defense can't key on one ball carrier.

The wildcat's ancestor is the single wing, the direct-snap formation that ruled football before the T-formation. Its modern NFL moment came in 2008, when the Miami Dolphins used it to stun the New England Patriots (the story is in the film room below), and for a season or two teams all over the league copied it. Then it faded, for a reason worth remembering: the player taking the snap usually couldn't throw. Once defenses stopped respecting the pass, they put an extra player in the box, and the wildcat's one-player advantage disappeared. It survives today in short bursts, as a short-yardage or goal-line package, or built around a player who really can both run and throw.

### Unbalanced line: tackle over

An **unbalanced line** puts more offensive linemen on one side of the center than the other. The common way, **tackle over**, moves one tackle to line up next to the other one, so one side has three linemen outside the center and the other side has only a guard.

## Balanced



## Tackle over

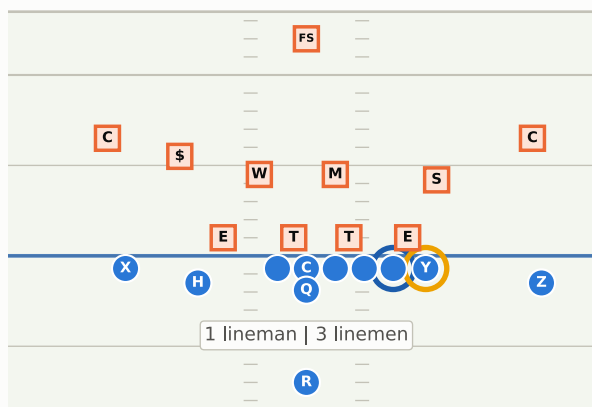


Figure 14: Balanced and unbalanced, against the same defensive front. Left: a normal line, two linemen on each side of the center, with the tight end (Y) on the right. Right: tackle over. The left tackle (ringed) lines up outside the right tackle, so the right side has three linemen plus the tight end, and the left side has only the guard. X lines up on the line on the left to keep seven on the line; he is now the left end. If the defense doesn't shift its front toward the extra lineman, the offense has an extra blocker on that side.

The point is arithmetic. A defense lines up its front by counting blockers on each side of the center. If one side suddenly has an extra lineman and the defense doesn't notice or doesn't shift, the offense has more blockers than defenders on that side, and it runs there. If the defense does shift, it has to shift its whole front a gap over, which can open a hole somewhere else or leave a pass-rusher out of position.

Two rules make the look possible and keep it honest. The five linemen wear numbers 50 to 79, which makes them ineligible to catch passes, so a tackle who moves to the end of the line is still just a blocker. That's why a tight end or receiver usually lines up on the line outside him: someone at each end has to be eligible if the offense wants the formation to threaten a pass to that side. And because the formation still needs seven players on the line, the side with only a guard needs a receiver on the line at its end, like X in the picture. (A lineman can become eligible, but only by reporting to the referee first, which is a different trick.)

## Swinging gate: make the defense follow

The **swinging gate** is a trick alignment you'll see mostly on extra points and two-point tries. The snapper stays over the ball, with the holder behind him, and most of the other linemen line up in a group far out to one side.

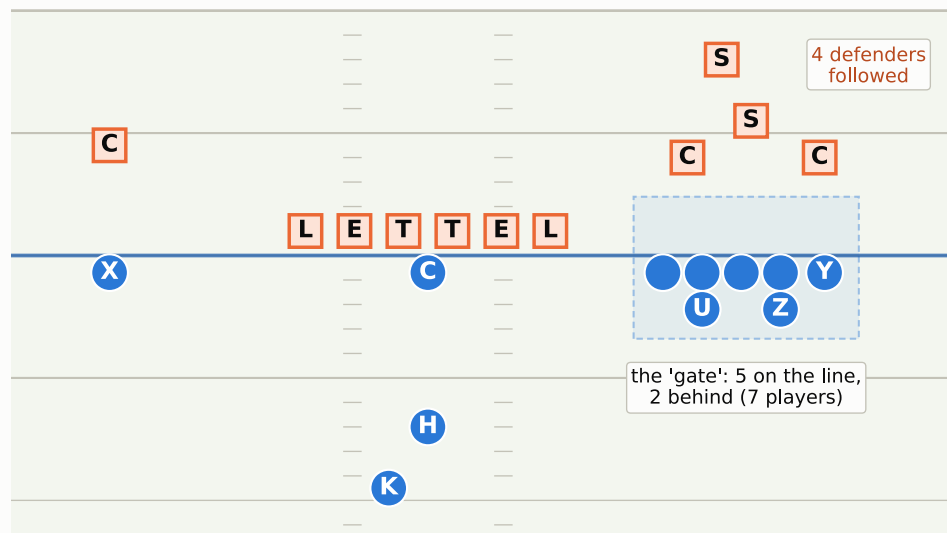


Figure 15: A swinging gate on a try after a touchdown. Only the snapper (C) is over the ball, with the holder (H) and kicker (K) behind him. Four linemen and a tight end stand on the line far to the right, with a second tight end (U) and a receiver (Z) behind them: the ‘gate,’ seven players in all. A receiver (X) on the line on the far left gives the formation its seventh man on the line, so it’s legal. Here the defense has sent only four players out to the gate, so the offense can snap it and throw or run to that side; if more defenders follow, the offense shifts back into its normal kicking formation (the gate ‘swings shut’).

The gate asks the defense a yes-or-no question. If the defense lines up to block the kick and ignores the group out wide, the offense snaps the ball to the holder, who throws to or pitches to the gate, where there are more offensive players than defenders. If the defense sends people out to cover the gate, the gate “swings shut”: the linemen run back to their normal spots and the team kicks, or the offense uses the defense’s confusion to run a play from the normal formation. Either way the defense has to recognize it, count it, and adjust in a few seconds, which is exactly the kind of pressure every formation in this chapter is designed to create.<sup>5</sup>

## Where spread formations came from

This chapter’s formations have a long history, and the full story belongs to Part 11. Here’s the short version, because it explains why the NFL looks the way it does now.

Spreading receivers out to stretch a defense is almost as old as the forward pass. In the 1930s and 1940s, Dutch Meyer’s TCU teams lined up receivers across the field in what Meyer called the “spread formation,” and in 1952 he wrote a book by that name, *Spread Formation Football*. In 1958, Glenn “Tiger” Ellison, a high school coach in Middletown, Ohio, whose team had started the season 0–4–1, tried a desperate formation with its receivers spread wide that his line coach nicknamed the “Lonesome Polecat.” Over the next seasons it grew into an offense of spread receivers whose routes adjusted to the defense on the run, the run and shoot. Darrel “Mouse” Davis refined it at Portland State in the 1970s, and by the early 1990s the Houston Oilers, Detroit Lions and Atlanta Falcons all ran it as their base offense. In the 1990s, Hal Mumme and Mike Leach’s Air Raid spread the field with four and five receivers at small colleges and then at Kentucky, and a generation of college offenses, passing and running alike, grew out of the same idea.<sup>6</sup>

The NFL came to spread formations slowly, and then all at once. One reason was supply: by the 2010s most of the quarterbacks and receivers arriving from college had played nearly every snap in the shotgun with three or four receivers spread out. Another was a demonstration. In 2007, the New England Patriots spread the field with Randy Moss and Wes Welker and produced the most prolific offense the league had seen to that point (see the film room). A decade later the league’s default formation was a shotgun

<sup>5</sup>The NFL moved the extra-point kick back to a snap from the 15-yard line in May 2015 (a 30–2 owners’ vote); two-point tries stayed at the 2. Gregg Rosenthal, “NFL moves extra point to 15-yard line for 2015 season,” NFL.com, May 19, 2015: <https://www.nfl.com/news/nfl-moves-extra-point-to-15-yard-line-for-2015-season-0ap3000000493347>. Both are unchanged for 2026.

<sup>6</sup>Meyer (TCU head coach 1934–1952) and *Spread Formation Football* (1952): “Dutch Meyer,” Wikipedia, [https://en.wikipedia.org/wiki/Dutch\\_Meyer](https://en.wikipedia.org/wiki/Dutch_Meyer), and “Spread offense,” [https://en.wikipedia.org/wiki/Spread\\_offense](https://en.wikipedia.org/wiki/Spread_offense). Ellison’s 0–4–1 start in 1958 and the “Lonesome Polecat” (named by line coach Stan Lewis), from Ellison’s own *Run and Shoot Football: Offense of the Future*, as summarized by Dusty Evely, “Tiger Ellison and the Run and Shoot”: <https://dustyevely.substack.com/p/tiger-ellison-and-the-run-and-shoot>. Davis at Portland State and the NFL users (Oilers, Lions, Falcons): “Run and shoot offense” and “Mouse Davis,” Wikipedia, [https://en.wikipedia.org/wiki/Run\\_and\\_shoot\\_offense](https://en.wikipedia.org/wiki/Run_and_shoot_offense), [https://en.wikipedia.org/wiki/Mouse\\_Davis](https://en.wikipedia.org/wiki/Mouse_Davis). Mumme and Leach at Iowa Wesleyan (1989–1991), Valdosta State (1992–1996) and Kentucky (1997–1998): “Mike Leach (American football coach),” Wikipedia, [https://en.wikipedia.org/wiki/Mike\\_Leach\\_\(American\\_football\\_coach\)](https://en.wikipedia.org/wiki/Mike_Leach_(American_football_coach)). For the wider story: Chris B. Brown, *The Art of Smart Football* (2015).

11-personnel spread, and the next innovation, the condensed formation, was a reaction to how well defenses had learned to play it. Each side's move invites the other's. You'll see that pattern in every part of this book; The Modern Era tells it in full.

## Film room

### i Film room: New England Patriots 2007

The 2007 Patriots went 16–0 in the regular season and scored 589 points, at the time an NFL record. Tom Brady threw 50 touchdown passes (also a record then), Randy Moss caught 23 of them (a single-season record), and Wes Welker caught 112 passes.<sup>7</sup> The offense, coordinated by Josh McDaniels, lived in spread formations: by the play-by-play data's shotgun flag, the Patriots were in the shotgun on about half their plays, more than any other team and nearly twice the league rate of about 27%.

The spread was built around two very different receivers, and it shows the two halves of this chapter. **Moss** was the deep threat outside, exactly the kind of receiver a 3x1 leaves on the island: a defense that rotated its safety away from him left him one-on-one with a cornerback, and a defense that kept a safety over him was short-handed everywhere else. **Welker** worked from the slot, where he had a two-way go on most snaps and could break inside or outside depending on how the defender played him.<sup>8</sup>

**What to notice:** when an offense puts one receiver alone on one side and three on the other, find the safeties. Where they go before the snap tells you which bet the defense is making, and which receiver the quarterback will look at first.

### i Film room: Los Angeles Rams 2018

Sean McVay's 2018 Rams went 13–3 and reached Super Bowl LIII, and as Personnel showed, they used 11 personnel on about 93% of their plays. With the personnel never changing, the variety had to come from formation and motion, and the Rams' signature was the **condensed formation**: receivers lined up close to the formation, so tight that about 60% of the Rams' formations were less than 20 yards wide at the snap, against 37% for the next-tightest team.

The tight splits made the run game and the passing game look alike. On outside zone runs, the receivers were close enough to block the linebackers and safeties at the edge; on play-action and bootlegs, the same receivers released into routes from the same spots, and defenders who had been hit by those receivers on the previous run had to guess.<sup>9</sup>

**What to notice:** before the snap, measure the outside receiver against the numbers painted on the field. If he's well inside them, look for a crack block on a run or an outside-breaking route on a pass, and watch the cornerback: if he widens to protect the outside, the inside is open.

<sup>7</sup>Pro Football Reference, 2007 New England Patriots: <https://www.pro-football-reference.com/teams/nwe/2007.htm> (16–0; 589 points; Brady 50 TD passes; Moss 23 TD receptions; Welker 112 receptions). The 2013 Denver Broncos later scored 606 points, and Peyton Manning threw 55 touchdown passes the same season. The Patriots lost Super Bowl XLII to the New York Giants, 17–14. Shotgun shares are a course calculation from nflverse play-by-play (shotgun flag, 2007 regular-season runs and passes): New England 50.3%, first in the league; league 26.9%.

<sup>8</sup>Welker's slot role and the three- and four-receiver sets: "The Near-Perfect Football Team," The Ringer, Aug. 7, 2017: <https://www.theringer.com/nfl/2017/8/7/16107814/2007-new-england-patriots-tom-brady-randy-moss-wes-welker>, which also describes the safety dilemma Moss created. How often Moss lined up alone on one side isn't recorded in the public data, so the text describes the dilemma rather than a frequency.

<sup>9</sup>Pro Football Reference, 2018 Los Angeles Rams: <https://www.pro-football-reference.com/teams/ram/2018.htm> (13–3; lost Super Bowl LIII to New England, 13–3). The 11-personnel share is a course calculation from nflverse participation data (see Personnel). Formation widths (ESPN Analytics and Next Gen Stats, through the 2018 season) and the condensed sets' uses, outside runs, receivers blocking and receivers releasing on bootlegs: Domonique Foxworth, "Yes, Sean McVay is an offensive genius," Andscape, Jan. 29, 2019: <https://andscape.com/features/rams-coach-sean-mcvay-is-an-offensive-genius>.

### i Film room: Miami Dolphins 2008, the Wildcat

On September 21, 2008, the Miami Dolphins, who had gone 1–15 the year before, traveled to New England and beat the Patriots 38–13. Running back Ronnie Brown took direct snaps in a formation the Dolphins called the Wildcat and scored four rushing touchdowns, and threw a touchdown pass from it as well. The loss ended New England’s 21-game regular-season winning streak, and Miami went on to win the AFC East at 11–5.<sup>10</sup>

The idea came to Miami through quarterbacks coach David Lee, who had run a version of it at the University of Arkansas with Darren McFadden. The name has a longer and disputed history: several high school and college coaches ran direct-snap formations under the same name in the late 1990s and 2000s, and all of them were rediscovering the single wing.<sup>11</sup>

**What to notice:** when a running back walks up to the shotgun spot and the quarterback jogs out wide, count the box. The wildcat’s edge is one extra runner; if the defense has one extra defender inside, it has taken that edge away, and the offense’s best play may be to shift back out of it.

### i Film room: Buffalo Bills 2019–2025, empty with Josh Allen

The Bills under Josh Allen have been one of the league’s steadier empty-backfield teams: about 10% of their plays from 2019 to 2025, against a league rate of about 8%, and in the top ten in six of those seven seasons. What makes their empty different is the quarterback.

Empty normally means “pass”: the league threw on more than 94% of its empty plays. But when the quarterback is a 6-foot-5, 237-pound runner, empty also creates a run: five receivers spread wide pull defenders out of the box, and if the defense matches with defensive backs and keeps only five men inside, the five offensive linemen can block all of them and the quarterback can run a draw (a run dressed up as a pass: he drops back, then takes off) at the safeties. In 2023, about one in eight of Buffalo’s empty snaps was a designed run, nearly all of them Allen, more than twice the league rate.<sup>12</sup>

**What to notice:** a running quarterback changes the arithmetic of every formation in this chapter, most of all empty. If the defense has only five in the box against an empty set, count the quarterback as a sixth runner.

### i Film room: Kansas City Chiefs, bunch and stack

Andy Reid’s Chiefs are known for stacking and bunching their receivers in the Patrick Mahomes era, often with Travis Kelce in the cluster, and film writers have pointed to those sets as a favorite answer to man coverage, Super Bowl LVII against Philadelphia among them.<sup>13</sup> The clusters do what this chapter describes: they give the receivers free releases against press, make man defenders work through traffic, and force the defense to agree on who takes whom at the snap.

**What to notice:** when three receivers line up in a cluster, watch the defenders over them at the snap. If each one chases his own man through the traffic, it’s man coverage, and the rubs are coming. If they stay in place and pass the receivers off, it’s a zone or a “switch” call, and the offense will look elsewhere.

<sup>10</sup>Pro Football Reference box score, Miami 38, New England 13 (Sept. 21, 2008): <https://www.pro-football-reference.com/boxscores/200809210nwe.htm>. Brown’s four rushing touchdowns and touchdown pass (19 yards to Anthony Fasano), the end of New England’s 21-game regular-season winning streak, Miami’s 1–15 2007 and 11–5 AFC East title in 2008, and David Lee’s Arkansas version with Darren McFadden and Felix Jones: ESPN, “Wildcat strike: How 2008 Dolphins shocked Patriots and the NFL”: [https://www.espn.com/blog/miami-dolphins/post/\\_id/28295/wildcat-strike-how-2008-dolphins-shocked-patriots-and-the-nfl](https://www.espn.com/blog/miami-dolphins/post/_id/28295/wildcat-strike-how-2008-dolphins-shocked-patriots-and-the-nfl); Pro Football Hall of Fame, “Wildcat immortalized”: <https://www.profootballhof.com/news/wildcat-immortalized>.

<sup>11</sup>“Wildcat formation,” Wikipedia: [https://en.wikipedia.org/wiki/Wildcat\\_formation](https://en.wikipedia.org/wiki/Wildcat_formation). The *Wall Street Journal* credited Hugh Wyatt, whose La Center (Washington) High School Wildcats ran a direct-snap version from 1997 and who described it in *Scholastic Coach* in 1998, with the name; Bill Snyder’s Kansas State Wildcats are also often cited as its namesake; Arkansas called its version the “Wild Hog.” Lee’s role in bringing it to Miami: the ESPN piece in the note above.

<sup>12</sup>Course calculation from nflverse data. Empty share: Next Gen Stats formation labels (2019–2022) and FTN backfield counts (2023–2025), regular-season runs and passes: Buffalo 10.0% of plays over 2019–2025, league 8.3%. Designed runs (runs that weren’t scrambles) as a share of empty snaps in 2023: Buffalo 12.7% (14 of 110, 13 by Allen), the rest of the league 5.3% (5.6% with Buffalo included). Allen’s listed height and weight: <https://www.buffalobills.com/team/players-roster/josh-allen/>.

<sup>13</sup>Barry Petchesky, Super Bowl LVII film breakdown, Defector, Feb. 13, 2023, on Kansas City’s stacked receivers in Super Bowl LVII and Andy Reid’s comment that the winning design worked against man coverage: <https://defector.com/chiefs-super-bowl-confuse-the-eagles-defense-with-this-one-weird-trick>.

## Watch for it on Sunday

💡 Watch for it: formation and leverage

- **Name the formation.** As the offense breaks the huddle, count receivers on each side, from the sideline in: 2x2, 3x1, empty (3x2), or a variation (bunch, stack, nub, quads). Is anybody beside the quarterback?
- **Find the strength.** Which side has more receivers? Is the tight end on the same side or the other one (a nub)? A split strength means the defense has to pick one.
- **Check the splits.** Are the outside receivers near the sidelines or inside the numbers? Condensed splits mean crack blocks, outside-breaking routes and rubs; wide splits mean the offense wants the defense spread thin.
- **Count each side.** Receivers versus defenders outside the box, side by side. Three receivers against two defenders is a quick throw waiting to happen; a lone receiver with no safety on his side is on an island.
- **Count the box.** Did the spread pull defenders out? Six or fewer in the box against a spread formation is an invitation to run.
- **Look for specialty clues.** The quarterback split out wide: wildcat. Three linemen on one side of the center: unbalanced. Linemen standing far out on an extra point: swinging gate.
- **Log it.** Add a “formation” column to your prediction log beside personnel: for example “11, 3x1 right, condensed.” After a half, check how often empty meant pass.

## Predict the play

💡 Predict the play 1: count the sides

Second-and-6 at midfield. The offense is in 11 personnel and lines up like this; the defense is in nickel with two deep safeties.

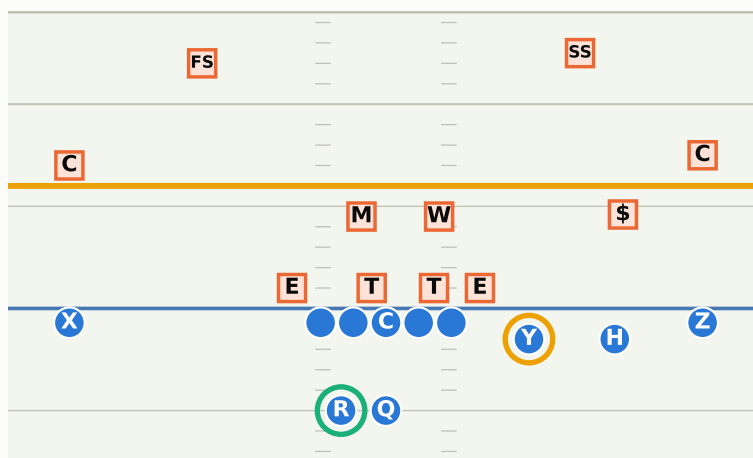


Figure 16: Drill 1. Defense: E = end, T = tackle, M and W = linebackers, \$ = nickel back, C = cornerbacks, FS and SS = safeties (both about 12 yards deep). Offense: X, H and Z are wide receivers, Y the tight end (gold ring), R the running back (aqua ring), Q the quarterback.

**Name the formation and number the receivers. Where does the offense have more receivers than defenders, and what's the likely play?**

*Answer 1 is at the end of the chapter. Try it first.*

💡 Predict the play 2: empty with a running quarterback

Third-and-4 at the opponent's 40. The offense has a quarterback who's one of the league's best runners. It goes empty, and the defense matches with six defensive backs.

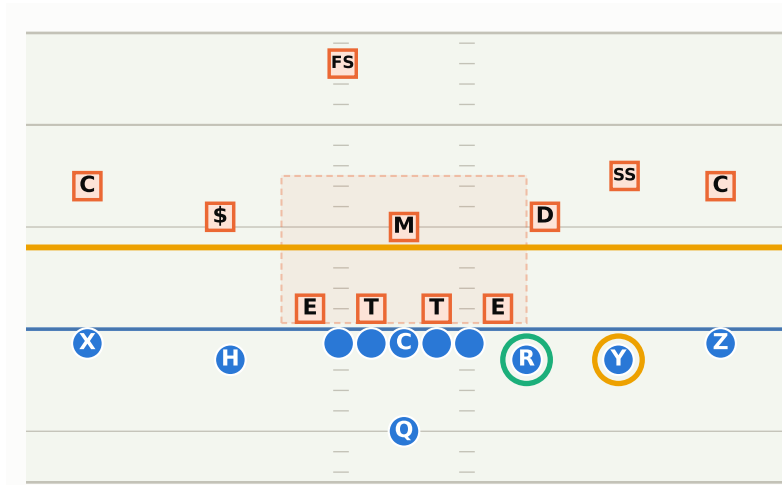


Figure 17: Drill 2. Empty 3x2 against a dime defense (six defensive backs). Defense: E = end, T = tackle, M = linebacker, C = cornerbacks, \$ = nickel, D = dime back, FS and SS = safeties. Shaded rectangle = the box.

**Is this a pass? What does the defense's alignment invite, and why?**

*Answer 2 is at the end of the chapter. Try it first.*

💡 Predict the play 3: everybody's tight

First-and-10 at the offense's own 35. The offense lines up in 11 personnel with the quarterback under center and the receivers squeezed in.

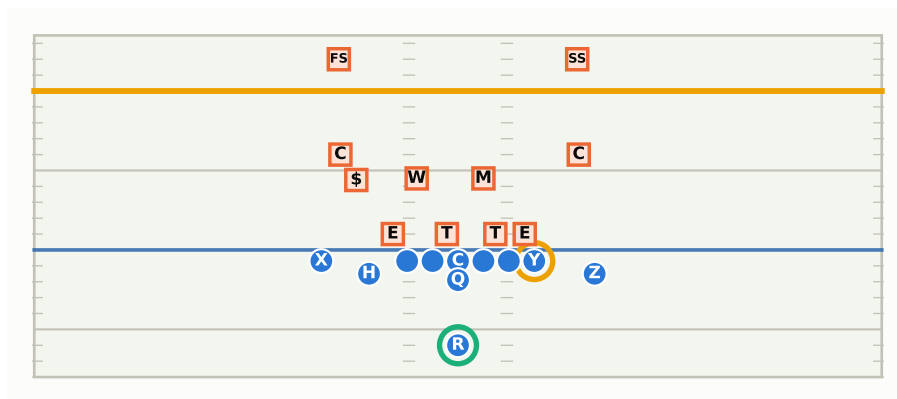


Figure 18: Drill 3. A condensed 2x2 with the ball in the middle of the field: X and Z about eight and a half yards from the ball, H and the tight end Y inside them. The defense has both cornerbacks (C) lined up inside the receivers, a nickel back (\$) and two deep safeties.

**Where is the open grass, and what two kinds of play does this formation set up? What would you expect the cornerbacks to do about it?**

*Answer 3 is at the end of the chapter. Try it first.*

## Takeaways

- **A formation is where the players stand, and a spread formation puts the pass-catchers across the width of the field.** It makes the defense defend all  $53\frac{1}{3}$  yards, thins the box and forces defenders to show where they're going. Personnel doesn't change; one grouping can line up a dozen ways.
- **Count receivers from the sideline in, side by side: #1, #2, #3.** That count names the formation (2x2, 3x1, empty 3x2, quads 4x1), sets the defense's coverage rules, and tells you where the offense has more receivers than the defense has defenders.
- **Every spread formation asks the defense a question it can't answer both ways.** Cover the slot or stay in the box; help the trips side or the lone receiver; match an empty set with defensive backs or keep a sixth man inside. The offense does whatever the defense didn't.
- **Clusters and lopsided formations have specific jobs.** Bunch and stack beat man coverage and press with traffic and free releases; nub splits the formation's strength; quads asks a question most defenses haven't practiced. Empty shows the quarterback everything but is more than 94% pass.
- **Condensed splits are the modern counter-move.** Pulling receivers in gives them room outside, a two-way go, angles for crack blocks and a way to make runs and passes look alike, at the cost of some width and a more crowded box.
- **Specialty looks (wildcat, unbalanced, swinging gate) are arithmetic tricks.** Each one adds a blocker or a runner somewhere the defense hasn't counted. Count the defense's answer before the snap.

## Connections

**Builds on:** Personnel (who is on the field; this chapter is where they stand), Formation Rules and the Language of Alignment (seven on the line, the X/Y/Z/H letters, splits and strength), and What a Play Really Is ("Gun Trips Right" as one chunk of a play call).

**Sets up:** A First Look at Defense (the box, the apex defender, man and zone, one-high and two-high: the defense's side of every picture in this chapter), Motion and Shifts as Weapons (moving between these formations before the snap to make the defense reveal itself), Passing Game Fundamentals (horizontal stretch and conflict defenders), The Quick Game (leverage and the two-way go in action), Dropback Concepts (rub routes from bunch and stack), Perimeter Runs and Misdirection (crack blocks and the crack toss), RPOs (the slot question turned into a play), and on defense, Disguise, Rotation, and Split-Field Coverage (how coverages adjust to 3x1 and trips checks).

**History and case studies:** The College Laboratory (where the spread offense grew up), The Modern Era (the spread, the condensed counter and the defenses in between), Belichick's Patriots: the Game-Plan Offense and Defense and the Shanahan tree.

### Go deeper

- Chris B. Brown, *The Art of Smart Football* (2015) and *The Essential Smart Football* (2012): short, clear essays on the spread, the run and shoot, the Air Raid and why formations create mismatches.
- Tim Layden, *Blood, Sweat and Chalk* (2010): the stories behind the single wing, the spread and the Wildcat, told through the coaches who built them.
- Pat Kirwan, *Take Your Eye Off the Ball* (2010; 2nd ed. 2015): a former NFL coach on reading formations and counting the defense before the snap.
- The 2025 NFL Big Data Bowl data on Kaggle includes a receiver-alignment label and full tracking for every play of the first nine weeks of the 2022 season. Measuring each team's receiver splits from the tracking coordinates is a good first project; the web edition of this chapter has starter code under "Show the Python" (not run when the book is built).

## Answers to Predict the play

### Answer 1.

**3x1 (trips right).** On the right, Z is #1, H is #2 and the tight end Y is #3; on the left, X is #1. The back beside the quarterback isn't numbered.

**Count the right side:** three receivers, but only two defenders outside the box over them, the cornerback on Z and the nickel on H. The third receiver, Y, has nobody within about seven yards: the nearest underneath defender is the linebacker W, who stayed inside the box, and the safety is twelve yards deep. That's three against two.

**The likely play:** a quick throw to the trips side, to Y or on a screen to the three-receiver side, before the defense can bring the safety down or widen W. If W walks out before the snap, the box drops to five and the run looks better: that's the slot question

from earlier in the chapter. Either way, the defense's alignment has handed the offense a numbers advantage, and good offenses check to it.

#### **Answer 2.**

**Probably a pass, but watch for a quarterback run.** League-wide, more than 94% of empty plays are dropbacks, and third-and-4 is a passing down. That's the default.

**What the alignment invites:** count the box. Four linemen and one linebacker: **five** defenders inside, against **five** offensive linemen. Every box defender has a blocker. The other six defenders are spread out over the five receivers, with the nearest safety twelve or thirteen yards deep. If the quarterback is a real running threat, a quarterback draw (he drops back as if to pass, then runs up the middle) has the numbers: five blockers on five defenders, and the quarterback against defensive backs who are backing up to cover receivers. That's the Bills' empty package with Josh Allen.

**The defense's dilemma:** to take the draw away it needs a sixth defender in the box, and the only place to get one is from coverage over five receivers. That's what empty is for.

#### **Answer 3.**

**The open grass is outside the receivers:** about 18 yards between each outside receiver and his sideline, and the cornerbacks, lined up *inside* them, aren't protecting any of it.

#### **Two kinds of play:**

1. **A run to the outside with crack blocks**, like the crack toss in this chapter. X and Z are already inside the nickel and the linebackers, so they can block down on them while a pulling lineman takes the cornerback, and the back has a lot of room outside.
2. **Outside-breaking routes:** corners, outs and fades into all that space. With the cornerback inside him, the receiver doesn't even need his two-way go; he has a one-way go to the side the defender isn't protecting.

**What the corners should do:** widen and play outside leverage, protecting the grass. That's the condensed formation's whole point. As soon as the corners widen, the inside opens for slants and digs, and the corners are now outside the receivers, where a crack block can't reach them, which leaves the crack-and-replace version of the toss (a pulling lineman on the corner) as the run to call. The defense can take away one thing; the formation has set up the other.